



Estimating NMR T_2 distribution data from well log data with the use of a committee machine approach: A case study from the Asmari formation in the Zagros Basin, Iran

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ABSTRACT

The Nuclear Magnetic Resonance (NMR) log is one of the most valuable logs in petroleum exploration which is used to precisely evaluate the reservoir and non-reservoir horizons. Along with porosity logs (neutron, density, sonic), NMR log is used to estimate the porosity and permeability of the hydrocarbon bearing intervals. The current study focuses on estimating NMR T_2 distribution data from conventional well log data with the use of artificial intelligent systems. The eight bin porosities of the combinable magnetic resonance (CMR) T_2 distribution alongside with the T_2 logarithmic mean (T_2 LM) values are predicted using the intelligent models developed in this study. The methodology applied here combines the results of the individual models in a committee machine with intelligent systems (CMIS) for estimating the NMR T_2 distribution and T_2 logarithmic mean data. The Fuzzy logic (FL), the adaptive neuro fuzzy system (ANFIS) and artificial neural networks (ANNs) are utilized as intelligent experts of the CMIS. The NN models are developed with four different training algorithms (Levenberg–Marquardt (LM), scaled conjugate gradient (SCG), one step secant (OSS) and resilient back-propagation (RP)) and the best one is chosen as the optimal NN expert of the CMIS. The CMIS assigns a weight factor to each individual expert by the simple averaging and weighted averaging methods. A genetic algorithm (GA) optimization technique is used to derive the weighted averaging coefficients. The results indicate that the GA optimized CMIS performs better than the individual experts acting alone for synthesizing the NMR T_2 curve and T_2 LM data from one specific set of conventional well logs.

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1. Introduction

NMR log provides an improved method to determine the reservoir petrophysical parameters. As an example since the NMR porosity is not affected by lithology and underground radioactive materials, it provides meticulous information about the porosity and pore size distribution of rocks. NMR logs have the possibility of in-situ measurement of the permeability, free fluid index, irreducible water and irreducible oil saturation. Moreover, identification of the fluid types and volumes is possible via the NMR T_2 data interpretation. Nevertheless, running NMR log in producing cased wells is not possible. On the other hand, it is one of the most expensive tools in the logging industry and its associated costs are the major limitation of its usage. Therefore,

there are only a few wells for which this log is run. In opposition to such problems, artificial intelligence has proved to be a cost-effective and time saving method to virtually extract the NMR log parameters. Recent studies have investigated the use of intelligent estimators such as artificial neural networks, fuzzy and adaptive neuro-fuzzy inference systems and the intelligent optimizers such as genetic algorithms to estimate the NMR log parameters from conventional well log data in many different cases.

Wong et al. (1995, 1998) used backpropagation neural networks to predict porosity and permeability of the oil wells. Mohaghegh et al. (1996) used ANNs for petroleum reservoir characterization. Mohaghegh et al. (2001) applied intelligent systems to synthesize artificial NMR logs in a field located in East Texas. Ogilvie et al. (2002) used GA and FL techniques to predict permeability from the NMR tool data. Chen and Lin (2006) used a committee machine with empirical formulas (CMEFs) for permeability prediction. Kadkhodaie-Ilkhchi et al. (2008, 2009) developed committee machines for prediction of total organic carbon and oil content from the well log data in the South Pars Gas Field,

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Iran. Rolon et al. (2009) used artificial neural networks to synthesize specific petrophysical logs from other logs.

Most of the previous works are in relation to estimating the parameters derived from NMR log (e.g. porosity and permeability). The current study focuses on estimating the original NMR T_2 and T_2 LM distribution alongside with the CMR bin porosities. Since T_2 and other NMR logs can be in relation to many other petrophysical and geological properties (e.g. facies, rock types, pore throat network, fluid saturation, wettability) it is worth synthesizing virtual NMR logs directly from one specific set of conventional well log data which are available for almost all wells. This study suggests that using the different neural network training algorithms and combining the best NN model with the FL and NF models can lead to the high accuracy of the results for construction of the synthetic T_2 curves.

The study area is located in Zagros basin, western Iran. The hydrocarbon bearing units of this area are Pabdeh, Asmari, Gajbaran, Agajari, etc. In the area that this research was carried out, the Paleocene to Pliocene Asmari formation is the most important reservoir unit that mainly consists of limestone, dolomite, shaly limestone and shale. The low-porosity and high-permeability Asmari formation undergoes minor lithological changes throughout the area. The latter fact provides capability of employing developed models of the current research in the neighboring fields. It is assumed that the formation contains no ions of chromium, manganese, iron, nickel or other paramagnetic ions and also no iron containing minerals which could considerably influence NMR measurements. Drilling fluid did not contain these ions neither.

2. NMR log

The nuclear magnetic resonance log is used as a tool to study the structure of porous media and their various rock–fluid interactions. This technique allows the determination of rock and fluid properties such as porosity and pore size distribution, permeability, water saturation, wettability, viscosity, etc. NMR logging exploits the large magnetic moment of hydrogen, which is abundant in rocks in the form of oil, gas and water. The NMR signal amplitude is proportional to the quantity of hydrogen nuclei present in a formation and can be calibrated to give a value for porosity that is free from lithology effects (Allen et al., 1997). The most important mechanism affecting the NMR relaxation is the grain-surface relaxation. Molecules in fluids are in constant Brownian motion, diffusing about the pore space and bouncing off the grain surfaces. Upon interaction with the grain surface, hydrogen protons can transfer some nuclear spin energy to the grain (contributing to T_1 relaxation) or irreversibly dephase (contributing to T_2 relaxation). Therefore the speed of relaxation most significantly depends on how often the hydrogen nuclei collide with the grain surface and this is controlled mainly by the surface-to-volume ratio of the pore in which the nuclei are located. Collisions are less frequent in larger pores resulting in a slower decay of the NMR signal amplitude and allowing a petrophysicist to understand the distribution of pore sizes (Kenyon et al., 1995).

3. Theory and methodology

The methodology applied in this work is based on selection of the appropriate input data. This step leads to selection of six well logs as the input data. The selection process will be explained in detail in the forthcoming section. The second step is to develop the different fuzzy, neuro-fuzzy and NN models. The NN models were developed with the four training algorithms including Levenberg–

Marquardt (Hagan and Menhaj, 1994), Resilient Back-Propagation (Riedmiller, 1993), Scaled Conjugate Gradient (Moller, 1993) and One Step Secant (Battiti, 1992). The next step is selection of the

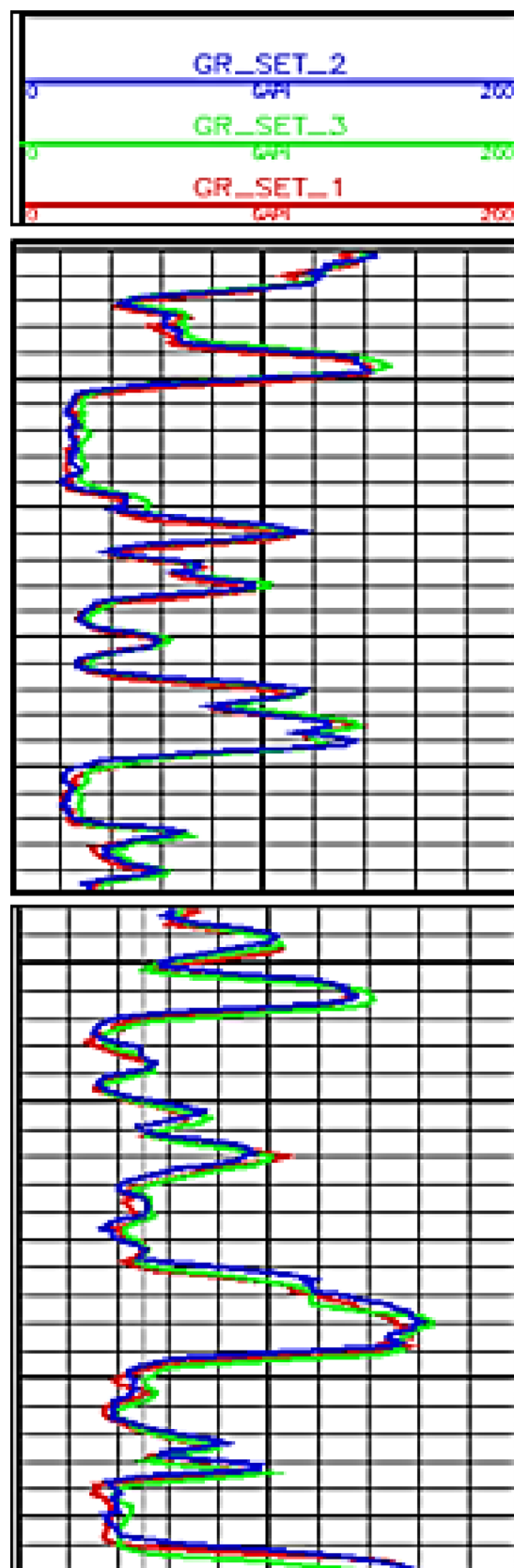


Fig. 1. GR logs from three sets of raw data used as depth controllers.

best FL, NF and NN models among the numerous intelligent models. The selection procedure was based on the minimum prediction error. The fourth step is to construct the CMIS. The individual experts of the CMIS were combined by the simple and weighted averaging methods. The optimal weigh values for the weighted averaging CMIS were extracted by the genetic algorithm optimization technique.

Artificial Neural Networks are inspired by biological nervous systems and are composed of parallel operating elements. These networks are trained for function fitting, pattern recognition and data clustering. Back propagation Neural Network (BP-NN) is a type of supervised-training networks. A three layered network with a tangent sigmoid (TANSIG) transfer function in hidden layer and a linear transfer function (PURELIN) in output layer was satisfactory for the aim of the present research. This kind of network is capable of fitting approximately any practical function having a finite number of discontinuities (Matlab user's guide,

2008). In this study four different BP-NNs with four neurons in hidden layer but Levenberg–Marquardt (LM), scaled conjugate gradient (SCG), one step secant (OSS) and resilient back-propagation (RP) training algorithms were utilized and the network with the best performance (with minimum error) was selected as the final NN model for estimating T_2 distribution data.

Fuzzy sets were introduced by Zadeh (1965) to treat data that were rather fuzzy than precise. It maps an input space to an output space through a fuzzy inference system (FIS). Two important types of fuzzy inference systems are the Mamdani-type FIS and Takagi-Sugeno-type FIS (Matlab user's guide, 2008). The Sugeno-type model which is implemented in this study uses the subtractive clustering in the fuzzy inference engine. The subtractive clustering is an effective, robust and quick approach to estimate the number of fuzzy clusters and cluster centers (Jarrah and Halawani, 2001).

Neural networks have proved to act well in a wide variety of applications but they cannot explain the rules and decisions they

Table 1

Correlation coefficients between CBPs and available well log data.

	CBP1	CBP2	CBP3	CBP4	CBP5	CBP6	CBP7	CBP8	T_2 LM
NPHI (pu)	0.0549	0.095	0.3113	0.5006	0.3589	0.15	0.4037	0.4913	0.1648
RHOZ (kg/m ³)	5E – 5	0.0073	0.4763	0.4968	0.4576	0.635	0.7585	0.468	0.1234
SPHI (pu)	0.0267	0.0679	0.4802	0.5111	0.4435	0.4012	0.7167	0.4538	0.1091
GR (GAPI)	0.5558	0.7103	0.0492	0.0142	0.0446	0.1564	0.0991	0.016	0.0323
PEFZ (barns/electron)	0.0062	0.0222	0.4073	0.3307	0.3109	0.5413	0.595	0.3108	0.0482
PHIE (pu)	0.0797	0.0593	0.3207	0.6734	0.66	0.8164	0.8681	0.6686	0.2457

Table 2

Parameters of input membership functions for fuzzy models.

	NPHI		RHOZ		SPHI		GR		PEFZ		PHIE	
	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ
CBP1												
mf1	0.1129	0.1961	267.2000	2290.0000	0.1012	0.1886	51.6900	33.5200	1.2540	1.9880	0.0899	0.2178
mf2	0.1129	0.1168	267.2000	2719.0000	0.1012	–0.0188	51.6900	39.4500	1.2540	3.8150	0.0899	0.0714
mf3	0.1129	0.0021	267.2000	3007.0000	0.1012	–0.0624	51.6900	2.6350	1.2540	5.4490	0.0899	0.0000
CBP3												
mf1	0.0627	0.0875	144.6000	2740.0000	0.0577	–0.0389	28.6300	34.2600	0.6577	3.8820	0.0488	0.0577
mf2	0.0627	0.1901	144.6000	2266.0000	0.0577	0.1981	28.6300	28.5300	0.6577	1.8430	0.0488	0.2357
mf3	0.0627	0.0012	144.6000	2926.0000	0.0577	–0.0572	28.6300	3.3640	0.6577	5.2900	0.0488	0.0000
mf4	0.0627	0.1294	144.6000	2583.0000	0.0577	0.0387	28.6300	30.2100	0.6577	4.4890	0.0488	0.1020
mf5	0.0627	0.2248	144.6000	2539.0000	0.0577	0.0847	28.6300	36.6400	0.6577	3.5560	0.0488	0.1620
mf6	0.0627	0.2080	144.6000	2313.0000	0.0577	0.1971	28.6300	81.4500	0.6577	2.3600	0.0488	0.1046
CBP5												
mf1	0.1127	0.1147	257.9000	2738.0000	0.1005	–0.0184	50.7900	46.6800	1.1820	3.9410	0.0869	0.0669
mf2	0.1127	0.1927	257.9000	2302.0000	0.1005	0.1936	50.7900	30.5500	1.1820	1.8750	0.0869	0.2135
mf3	0.1127	0.1763	257.9000	2407.0000	0.1005	0.1176	50.7900	37.9800	1.1820	4.6030	0.0869	0.1569
CBP7												
mf1	0.0439	0.1901	104.0000	2266.0000	0.0460	0.1981	21.8500	28.5300	0.5106	1.8430	0.0399	0.2357
mf2	0.0439	0.1133	104.0000	2752.0000	0.0460	–0.0273	21.8500	48.5200	0.5106	3.8810	0.0399	0.0584
mf3	0.0439	0.0501	104.0000	2664.0000	0.0460	–0.0290	21.8500	18.4300	0.5106	4.8540	0.0399	0.0465
mf4	0.0439	0.1454	104.0000	2489.0000	0.0460	0.0792	21.8500	31.3900	0.5106	4.4330	0.0399	0.1232
mf5	0.0439	0.1822	104.0000	2318.0000	0.0460	0.1710	21.8500	46.8000	0.5106	2.0670	0.0399	0.1713
mf6	0.0439	0.2030	104.0000	2543.0000	0.0460	0.0648	21.8500	29.4800	0.5106	3.4220	0.0399	0.1572
mf7	0.0439	0.0463	104.0000	2807.0000	0.0460	–0.0883	21.8500	17.3100	0.5106	3.9920	0.0399	0.0361
mf8	0.0439	0.1640	104.0000	2641.0000	0.0460	0.0186	21.8500	50.6600	0.5106	3.2310	0.0399	0.0944
T_2 LM												
mf1	0.0502	0.1867	120.5000	2267.0000	0.0469	0.1992	22.6100	29.9800	0.5262	1.8940	0.0413	0.2306
mf2	0.0502	0.0875	120.5000	2740.0000	0.0469	–0.0389	22.6100	34.2600	0.5262	3.8820	0.0413	0.0577
mf3	0.0502	0.1969	120.5000	2370.0000	0.0469	0.1904	22.6100	77.8100	0.5262	2.2480	0.0413	0.0981
mf4	0.0502	0.0364	120.5000	2614.0000	0.0469	–0.0426	22.6100	29.1100	0.5262	4.7600	0.0413	0.0424
mf5	0.0502	0.1921	120.5000	2591.0000	0.0469	0.0494	22.6100	34.0800	0.5262	3.7780	0.0413	0.1434
mf6	0.0502	0.0094	120.5000	2921.0000	0.0469	–0.0467	22.6100	12.1300	0.5262	5.1170	0.0413	0.0000
mf7	0.0502	0.1887	120.5000	2440.0000	0.0469	0.1746	22.6100	113.5000	0.5262	2.6330	0.0413	0.0270
mf8	0.0502	0.1279	120.5000	2592.0000	0.0469	0.0495	22.6100	32.2700	0.5262	4.9480	0.0413	0.0988

Table 3

Parameters of output membership functions for fuzzy models.

	C1	C2	C3	C4	C5	C6	C7
CBP1							
mf1	−0.0379	0.0000	0.0141	0.0004	0.0002	0.1031	−0.1025
mf2	0.0429	0.0000	−0.0079	0.0001	0.0007	−0.0331	0.0076
mf3	−0.2687	0.0000	0.0989	0.0000	−0.0075	0.3965	−0.0219
CBP3							
mf1	−0.0227	0.0000	0.0214	0.0000	−0.0049	−0.0361	0.0259
mf2	−0.0732	−0.0001	−0.0419	0.0000	−0.0073	−0.1090	0.2399
mf3	−0.0724	0.0000	−0.0123	0.0002	0.0009	−0.0757	0.0422
mf4	0.0715	0.0000	−0.1054	−0.0001	−0.0006	−0.1387	0.1425
mf5	0.0203	−0.0001	−0.0313	−0.0001	0.0037	−0.0410	0.2211
mf6	−0.0679	−0.0001	−0.0034	−0.0001	0.0033	−0.1300	0.2157
CBP5							
mf1	0.0200	0.0000	0.0082	0.0000	0.0008	0.0300	−0.0160
mf2	0.0339	−0.0001	−0.1033	−0.0002	−0.0008	−0.0035	0.1628
mf3	−0.0017	0.0000	−0.0178	0.0001	−0.0030	0.2296	0.0169
CBP7							
mf1	0.2083	0.0001	0.1424	0.0004	−0.0458	0.2271	−0.1290
mf2	−0.0070	0.0001	−0.0058	0.0001	0.0085	0.2039	−0.1696
mf3	0.1173	−0.0001	0.0873	−0.0001	0.0046	−0.0910	0.1413
mf4	0.0449	0.0000	0.0836	−0.0002	0.0092	−0.0369	0.0419
mf5	−0.1437	0.0000	−0.1321	0.0005	−0.0036	0.5821	0.0287
mf6	−0.2236	0.0000	−0.0175	0.0008	−0.0047	0.5912	−0.0418
mf7	−0.0541	0.0000	0.0129	0.0002	0.0035	0.1402	−0.1366
mf8	−0.1191	0.0000	−0.0094	0.0000	0.0084	0.0728	0.1057
T2LM							
mf1	−0.5638	−0.0015	−0.2887	−0.0050	0.4275	−1.2250	3.4400
mf2	1.2900	0.0005	−1.4410	0.0024	0.2664	3.4730	−2.7360
mf3	2.7260	−0.0011	−1.8600	−0.0095	0.1583	−1.7430	3.0430
mf4	1.3050	−0.0009	0.6618	0.0004	0.2102	−2.5090	1.5030
mf5	−0.8971	0.0001	1.0550	−0.0014	0.1461	0.3975	−0.4744
mf6	0.7154	0.0004	−0.2974	−0.0020	−0.1467	−0.0543	−0.5049
mf7	0.4377	−0.0003	−1.0790	0.0013	−0.0787	2.6550	0.8095
mf8	0.5518	−0.0015	2.2400	−0.0009	0.2585	−1.4580	2.9480

Table 4

Parameters of input membership functions for neuro–fuzzy models.

	NPHI		RHOZ		SPHI		GR		PEFZ		PHIE	
	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ
CBP1												
mf1	0.1334	0.2437	267.2000	2290.0000	0.0868	0.2432	51.6900	33.5200	1.2590	1.9900	0.1353	0.2250
mf2	−0.0829	0.1184	267.2000	2719.0000	−0.0667	0.1010	51.6900	39.4500	1.2620	3.8120	0.1155	0.0575
mf3	0.0937	0.0111	267.2000	3007.0000	0.0963	−0.0640	51.6900	2.6350	1.2630	5.4430	0.1216	0.0334
CBP3												
mf1	0.1095	0.0935	144.6000	2740.0000	0.0764	−0.0389	28.6300	34.2600	0.6593	3.8810	0.0486	0.0651
mf2	0.0733	0.1988	144.6000	2266.0000	0.0705	0.1869	28.6300	28.5300	0.6605	1.8450	0.1088	0.2091
mf3	0.0616	0.0027	144.6000	2926.0000	0.0529	−0.0529	28.6300	3.3640	0.6578	5.2900	0.0492	0.0016
mf4	0.1099	0.2040	144.6000	2583.0000	0.0441	0.1069	28.6300	30.2100	0.6662	4.4840	0.0640	0.1230
mf5	0.0724	0.2554	144.6000	2539.0000	0.0808	0.0529	28.6300	36.6400	0.6549	3.5590	0.0940	0.1373
mf6	0.0486	0.1970	144.6000	2313.0000	−0.0014	0.2350	28.6300	81.4500	0.6505	2.3560	0.0211	0.1072
CBP5												
mf1	0.0419	0.0908	200.6000	2738.0000	0.0340	−0.0183	39.5000	46.6800	0.9206	3.9450	0.0904	0.0625
mf2	0.1485	0.1999	200.6000	2287.0000	0.0940	0.1896	39.5000	31.0300	0.9160	1.8700	0.1259	0.1541
mf3	0.0532	0.1755	200.6000	2544.0000	0.0766	0.1095	39.5000	28.6300	0.9029	3.6090	0.1067	0.1450
mf4	0.1498	0.0653	200.6000	2884.0000	0.1056	0.0123	39.5000	13.5200	0.9251	5.1220	0.1249	0.0483
CBP7												
mf1	0.0553	0.0937	182.0000	2719.0000	0.0524	−0.0670	38.2400	39.4500	0.8908	3.8140	0.0374	0.0975
mf2	0.0716	0.2051	182.0000	2266.0000	0.0744	0.2139	38.2400	28.5300	0.8880	1.8410	0.0220	0.2508
mf3	0.0987	0.1481	182.0000	2457.0000	0.0777	0.0795	38.2400	36.5200	0.8947	4.3750	0.1403	0.0978
T2LM												
mf1	0.0417	0.1837	90.3900	2251.0000	0.0084	0.2232	16.9600	29.6200	0.3885	1.8240	0.0130	0.2562
mf2	0.1074	0.1968	90.3900	2285.0000	0.0428	0.1936	16.9600	50.8400	0.3708	2.0630	0.0362	0.1967
mf3	0.0366	0.0901	90.3900	2740.0000	0.0300	−0.0593	16.9600	34.2600	0.3935	3.8810	0.0665	0.0900
mf4	0.0721	0.2281	90.3900	2370.0000	0.0964	0.1719	16.9600	90.2900	0.4237	2.3530	0.0689	0.0951
mf5	0.0377	0.0068	90.3900	2965.0000	0.0366	−0.0522	16.9600	4.7300	0.3946	5.3570	0.0310	0.0000
mf6	0.0745	0.0433	90.3900	2634.0000	0.0677	−0.0317	16.9600	29.7900	0.3944	4.2290	0.0674	0.0512
mf7	0.0506	0.1196	90.3900	2770.0000	0.0491	−0.0153	16.9600	54.5600	0.3972	3.8020	0.0712	0.0382
mf8	0.0566	0.1969	90.3900	2440.0000	0.0514	0.1730	16.9600	113.5000	0.3868	2.6310	0.0484	0.0361

Table 5
Parameters of output membership functions for neuro-fuzzy models.

	C1	C2	C3	C4	C5	C6	C7
CBP1							
mf1	−0.0431	0.0001	0.0156	0.0005	−0.0015	0.1512	−0.1772
mf2	−0.0109	0.0000	−0.0072	0.0002	0.0012	0.0439	−0.0315
mf3	−0.0039	0.0000	0.0237	0.0000	−0.0010	−0.0077	0.0152
CBP3							
mf1	−0.0168	0.0000	0.0175	0.0000	−0.0008	0.0192	0.0159
mf2	−0.0764	0.0000	−0.0613	0.0002	−0.0044	−0.0029	0.1258
mf3	−0.1625	0.0000	−0.0203	0.0002	−0.0012	−0.0411	0.0233
mf4	0.0485	0.0000	0.0899	−0.0002	0.0113	−0.1467	0.0114
mf5	−0.0198	−0.0001	0.0266	−0.0002	0.0105	−0.0909	0.2142
mf6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CBP5							
mf1	0.0975	0.0000	−0.0055	−0.0001	0.0002	−0.0200	−0.0073
mf2	0.0096	0.0000	−0.0428	0.0000	−0.0002	0.0797	0.0580
mf3	0.0487	0.0000	−0.0008	0.0001	−0.0002	0.1900	0.0245
mf4	−0.1191	0.0000	0.0815	0.0001	0.0001	0.0939	−0.0225
CBP7							
mf1	0.0502	0.0000	0.0451	0.0001	0.0020	0.1505	−0.1391
mf2	0.0361	0.0000	−0.0114	0.0006	−0.0226	0.2510	−0.0371
mf3	−0.0205	0.0000	0.0335	−0.0001	−0.0026	0.2134	0.0697
T2LM							
mf1	0.9826	−0.0013	−12.5200	−0.0192	0.4061	6.5070	3.7730
mf2	2.1560	−0.0061	−2.4030	−0.0266	0.4250	−11.5800	16.9700
mf3	8.9070	0.0000	1.4990	−0.0022	0.2698	−7.0770	−0.8973
mf4	2.5240	−0.0029	−2.0150	−0.0162	0.1838	−10.5800	8.8640
mf5	−0.1289	0.0003	0.2836	−0.0028	−0.1275	5.0810	−0.0960
mf6	1.1890	−0.0007	1.6960	−0.0012	0.0968	−1.1920	1.5210
mf7	−2.1280	0.0007	−1.9050	0.0089	0.1577	9.3450	−3.0690
mf8	0.4120	−0.0003	−1.3740	−0.0023	0.0490	−2.3040	1.0420

Table 6
Parameters of the final fuzzy and neuro-fuzzy models.

FL models				NF models				
	r	No. of MFs	Type of MFs	r	No. of MFs	Type of MFs	Max. number of training epochs	The best model occurrence epoch
CBP1	0.9	3	Gaussian	0.9	3	Gaussian	200	9
CBP2	0.3	7	Gaussian	0.8	4	Gaussian	300	6
CBP3	0.5	6	Gaussian	0.5	6	Gaussian	300	4
CBP4	0.9	3	Gaussian	0.6	5	Gaussian	300	5
CBP5	0.9	3	Gaussian	0.7	4	Gaussian	500	7
CBP6	0.5	6	Gaussian	0.9	3	Gaussian	150	24
CBP7	0.4	8	Gaussian	0.7	3	Gaussian	250	3
CBP8	0.5	7	Gaussian	0.5	7	Gaussian	200	3
T ₂ LM	0.4	8	Gaussian	0.3	8	Gaussian	100	47

Table 7
No. of training data and the best training algorithm for neural network models.

	Total no. of training data points	No. of training data points	No. of validation data points	Training algorithm
CBP1	698	559	139	SCG
CBP2	732	586	146	RP
CBP3	668	535	133	LM
CBP4	702	562	140	SCG
CBP5	680	544	136	LM
CBP6	760	608	152	SCG
CBP7	728	583	145	OSS
CBP8	660	528	132	OSS
T ₂ LM	1069	856	213	SCG

make. Fuzzy logic is good at this case but it cannot automatically acquire the rules it uses to make those decisions. The ANFIS or adaptive neuro fuzzy inference system is a hybrid system which

combines the neural networks and fuzzy logic to overcome these problems. ANFIS allows the membership function parameters to be chosen in a way that the given input–output data is tracked in the best way. This is because these parameters are chosen so as to tailor the membership functions to the input/output data in order to account for the types of variations in the data values (Matlab user's guide, 2008). In this study 10 groups of NF models were constructed based on the ANFIS step sizes of 0.1–1. Among each group a single model was picked as the best and the final NF model was chosen from among these 10 models.

The genetic algorithm (GA) introduced by John Holland (1975) is an optimization and search technique based on the principles of genetics and natural selection (Haupt and Haupt, 1998). It starts with a number (population) of numerical arrays known as chromosomes (individuals) that represent possible answers of the problem. At the next step, the fitness value of these chromosomes is computed and the genetic operator (crossover or mutation) is

applied on those with better fitness values to produce new individuals (generation). GA is fundamentally based on the survival of the best chromosomes.

A committee machine is an ensemble of estimators consisting typically of neural networks or decision trees. It is helpful if mapping an input space to an output space is not approximated by a single estimator but by several estimators, where each estimator can focus on a particular region in the input space. Each individual estimator is called an expert. In this study, experts of the committee machine

were FL, BP-NN and NF. These experts could be combined by the simple averaging or weighted averaging methods. The different optimization algorithms such as the genetic algorithm and pattern search method could be used to determine the optimal contribution of weights.

4. Case study

4.1. Selection of appropriate input log data

The conventional well logs and NMR data used in this study as input–output data pairs came from one of the wells in the Zagros Basin penetrated into the Asmari Formation. Fortunately, the NMR log spanned over a large interval of about 345 m which provided enough measurements to develop and test the intelligent models in the same well. Approximately 235 m of this interval was used for training the intelligent models and the remaining 110 m-interval was used as unforeseen part of the well to evaluate the performance of the intelligent models. The reliability of the constructed models was measured on 78 testing depth samples. For the available data the first step was to process and remove the bad hole intervals. Gamma Ray log was employed as a depth

Table 8

Optimal CMIS weight values produced by genetic algorithm.

	NN model	FL model	NF model
CBP1	0.5964	0.4023	0.0004
CBP2	0.8380	0.0036	0.1584
CBP3	0.8216	0.1045	0.0739
CBP4	0.9500	1.521E–05	0.0490
CBP5	0.6674	0.0004	0.3330
CBP6	0.9878	0.0063	0.0069
CBP7	0.9975	3.715E–06	0.0015
CBP8	0.9990	5.376E–07	2.107E–07
T ₂ LM	0.7703	0.2287	9.815E–06

Table 9

Performance of intelligent models for synthesizing CBP1, CBP2 and CBP3.

Intelligent system	CBP1		CBP2		CBP3	
	MSE (pu) ²	R ²	MSE (pu) ²	R ²	MSE (pu) ²	R ²
FL	2.20E–05	0.7555	2.763E–05	0.7588	3.490E–05	0.5032
NF	3.42E–05	0.648	2.887E–05	0.7654	3.517E–05	0.4923
NN	2.09E–05	0.8117	1.890E–05	0.8054	3.207E–05	0.5350
GA Optimized CMIS	1.95E–05	0.8002	1.833E–05	0.8073	3.206E–05	0.5360
Simple Averaging CMIS	2.22E–05	0.7621	2.160E–05	0.7873	3.264E–05	0.5309

Table 10

Performance of intelligent models for synthesizing CBP4, CBP5 and CBP6.

Intelligent system	CBP4		CBP5		CBP6	
	MSE (pu) ²	R ²	MSE (pu) ²	R ²	MSE (pu) ²	R ²
FL	4.227E–05	0.6802	3.100E–05	0.8309	4.143E–05	0.9186
NF	3.685E–05	0.6901	3.030E–05	0.8226	3.701E–05	0.9456
NN	3.046E–05	0.8043	2.798E–05	0.8275	3.013E–05	0.9488
GA optimized CMIS	3.044E–05	0.8008	2.735E–05	0.8366	3.007E–05	0.9490
Simple averaging CMIS	3.456E–05	0.7405	2.834E–05	0.8388	3.400E–05	0.9483

Table 11

Performance of intelligent models for synthesizing CBP7, CBP8 and T₂LM.

Intelligent system	CBP7		CBP8		T ₂ LM	
	MSE (pu) ²	R ²	MSE (pu) ²	R ²	MSE	R ²
FL	2.270E–04	0.8691	1.666E–04	0.4822	1.025E–02	0.5788
NF	2.013E–04	0.8929	2.444E–04	0.3123	1.201E–02	0.5492
NN	1.612E–04	0.9075	9.964E–05	0.6422	7.560E–03	0.8777
GA optimized CMIS	1.606E–04	0.9075	9.945E–05	0.6422	7.270E–03	0.8430
Simple averaging CMIS	1.875E–04	0.9000	1.482E–04	0.5141	8.560E–03	0.7184

controlling tool as shown in Fig. 1. The gamma ray logs from the three different sets of raw data were used as depth controllers between the conventional and NMR logs. As known, the NMR T_2 distribution curves that produce observed magnetizations are correlated to measured spin echo trains through a mathematical inversion process. The echo trains are presented in the following equation.

$$\begin{aligned} \text{echo}(1) &= \phi_1 e^{-[t(1)/T_{2,1}]} + \phi_2 e^{-[t(1)/T_{2,2}]} + \phi_3 e^{-[t(1)/T_{2,3}]} + \dots + \phi_m e^{-[t(1)/T_{2,m}]} + \text{noise} \\ \text{echo}(2) &= \phi_1 e^{-[t(2)/T_{2,1}]} + \phi_2 e^{-[t(2)/T_{2,2}]} + \phi_3 e^{-[t(2)/T_{2,3}]} + \dots + \phi_m e^{-[t(2)/T_{2,m}]} + \text{noise} \\ &\vdots \\ \text{echo}(n) &= \phi_1 e^{-[t(n)/T_{2,1}]} + \phi_2 e^{-[t(n)/T_{2,2}]} + \phi_3 e^{-[t(n)/T_{2,3}]} + \dots + \phi_m e^{-[t(n)/T_{2,m}]} + \text{noise} \end{aligned}$$

where $t(i) = i$ TE and $i = 1, \dots, n$, is the time when the i th echo was acquired. In the inversion process, a set of n linear equations of m unknowns, $\phi_1, \dots, \phi_m$ are inverted to set of ϕ_i . The inversion process must involve regularization to stabilize the solution; therefore the set

of ϕ_i is not unique. The area under the curves that represents porosity is generally robust though (Coates et al., 1999). Hence an equivalent normalization approach for all ϕ_i s was employed in the current study. Even though the T_2 distribution of rocks is normally a continuous function, the mapping process uses a multi-exponential model that assumes that the T_2 distribution consists of m discrete relaxation times T_{2i} with corresponding porosity components ϕ_i (Coates et al., 1999). The NMR T_2 distribution data were divided into eight equally sized bins. These bins or T_2 arrays are referred to as CBP (Combinable Magnetic Resonance, CMR Bin Porosity) and are represented by CBP1 to CBP8, respectively. Each bin is referred to as the amount of porosity from the T_2 distribution within a specific time range. Developing a strong intelligent model capable of making accurate predictions vitally needs using noise-free input arguments, as a single noisy input value disturbs the weight matrix in neural networks or the fuzzy rules in fuzzy inference systems. With this in mind and the fact that simultaneous modeling of all CBPs proved to be displeasing, cross plot and regression analysis were implemented and narrowly

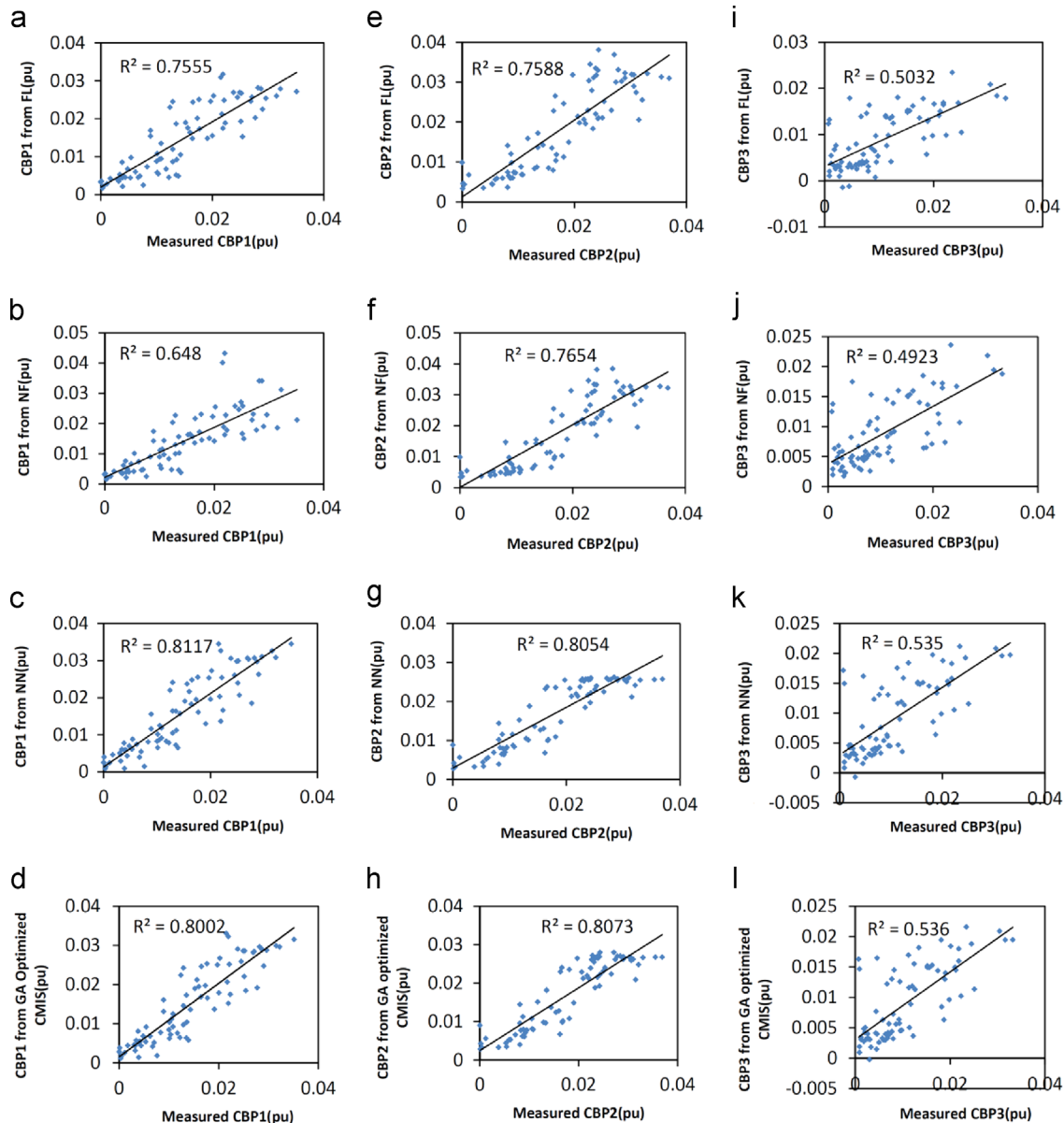


Fig. 2. Cross plots showing correlation coefficients between measured and predicted (a–d) CBP1, (e–h) CBP2 and (i–l) CBP3 values in test data.

distinguished input values were chosen to develop the intelligent model of each CBP independently. According to the cross plot and regression analysis, neutron porosity (NPHI), bulk density (RHOZ), sonic porosity (SPHI), gamma ray (GR), standard resolution photo-electric factor (PEFZ) and effective porosity data from the high resolution integrated logging tool (PHIEHILT) showed stronger relationships with CBPs. The relationships (correlation coefficients) between CBPs and available well log data are presented in Table 1. The stronger the relationships between input and target logs are, the more accurate the predictions are. Wide-range variations of the T_2LM values i.e. from 0.6151 to 586.8975 ms could cause lower performance of the constructed models; hence the T_2LM data were normalized between 0 and 1 before constructing models and

therefore the performance of the T_2LM models was represented in normal scale (i.e. without unit).

4.2. Fuzzy models

The fuzzy models used in this research are the Sugeno-type FIS developed in the programming environment of MATLAB software. The membership functions (MFs), their parameters and fuzzy “if-then” rules were determined by the subtractive clustering method. This sort of clustering offers the capability of tracking the input-output data in the best way. The clustering radius (r) was the major factor controlling the number of clusters, fuzzy rules and accuracy of the fuzzy models. The optimal FIS with minimum error

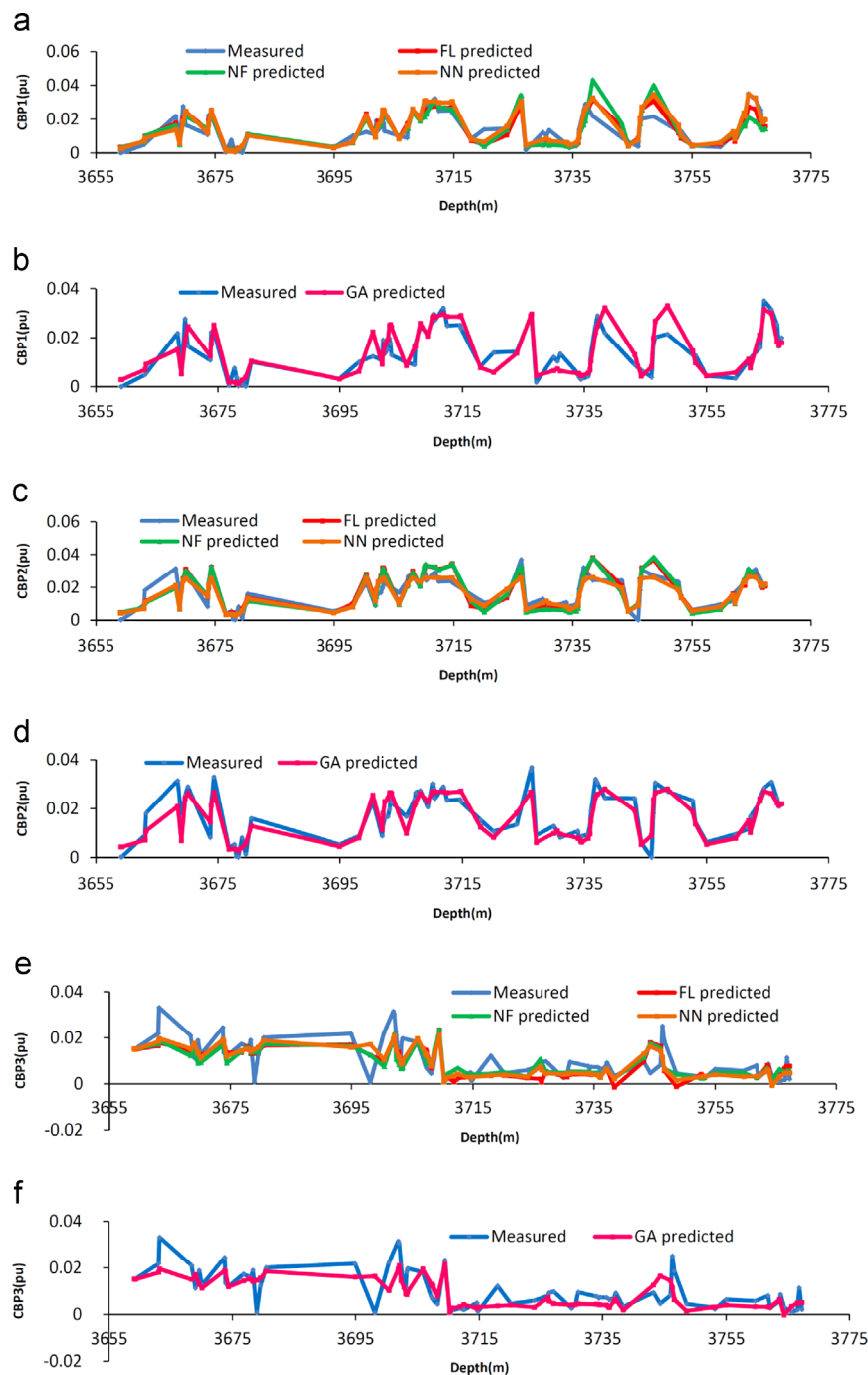


Fig. 3. Comparison between measured and predicted (a, b) CBP1, (c, d) CBP2 and (e, f) CBP3 values in test data versus depth.

was extracted by assigning a set of values between 0.1 and 1 to the clustering radius. The parameters of input and output membership functions (mf_1 , mf_2 , ..., mf_n) are shown in Tables 2 and 3, respectively.

4.3. Neuro-fuzzy models

The NF models are quite similar to the fuzzy model except for the parameters of the membership functions that are adjusted by a BP-NN. The proper step size values and number of training epochs were obtained using a trial and error method. A step size value of 0.1 led to satisfactory results in all cases. Models were trained with the use of back propagation neural network and least squares error algorithm. For all NF models, the number of training epochs was set to values greater than 100. For each case, the maximum limit of this parameter was selected and no further change in the validation error was observed. Among all models the one with the best performance was selected as the final NF model for that case. For all NF models the error tolerance was set to

0.1 and MSE (mean squared error) was used as the performance criterion. The parameters of the input and output membership functions are shown in Tables 4 and 5, respectively. The optimal values of the clustering radius, the number and type of the membership functions and training epochs are presented in Table 6.

4.4. Neural network models

Four three layered back propagation neural networks including LM, RP, SCG and OSS training algorithms with four neurons in their hidden layer and TANSIG-PURELIN transfer functions were designed. The best performing NN model was chosen as the final NN model for estimating the T_2 distribution and T_2 LM data. The number of training and validation data and the best performing training algorithm for each neural network model are presented in Table 7. About 80% of the total training data pairs were used for training the networks and the remaining 20% were used to validate them. Finally, 78 data points, selected from different

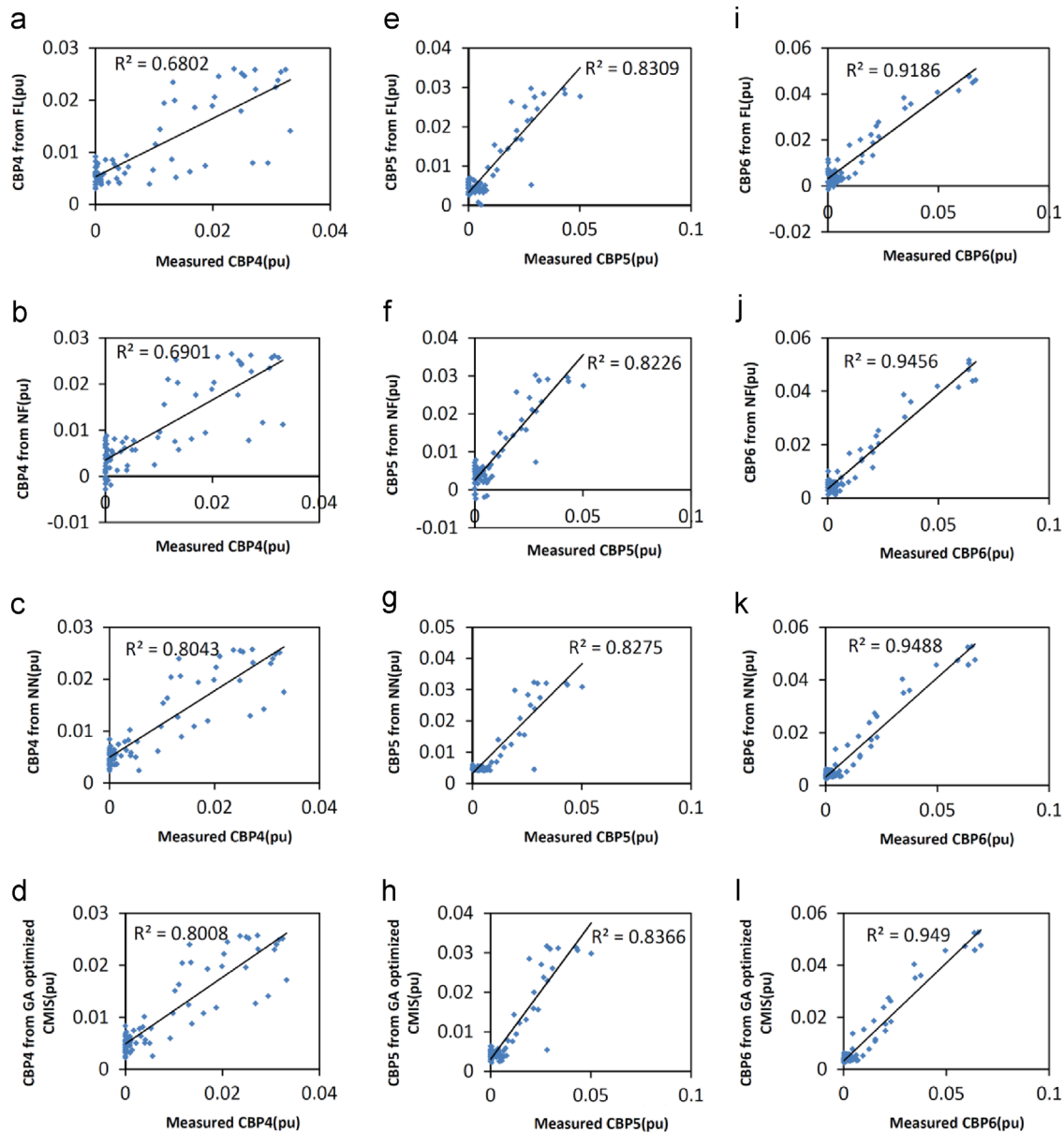


Fig. 4. Cross plots showing correlation coefficients between measured and predicted (a–d) CBP4, (e–h) CBP5 and (i–l) CBP6 values in test data.

lithologies inside the well were used to measure the reliability of the models. The results of neural network models for the CBPs and T₂LM data are presented in the coming sections.

4.5. CMIS models

The final FL, NF and NN models for CBPs and T₂LM data were combined in a CMIS through two different approaches. In the first approach, the equal weigh factors were assigned to each CMIS expert. In this method (known as the simple averaging CM), all experts have equal contribution in predicting the overall output. The final results of this approach were not satisfactory in comparison to the weighted averaging method. In the second approach, a genetic algorithm optimization technique was employed to obtain a better

combination of the individual experts. The optimized weight values led to the acceptable results and the constructed CMIS showed a better performance in comparison to the individual experts acting alone. For the genetic algorithm model, the initial population size was set to 30 and 20 for the CBPs and T₂LM data, respectively. The next generation was produced with a crossover fraction of 0.8 and elite count of 2. The optimized weight values for the CBPs and T₂LM models are presented in Table 8. The objective function that should be optimized with GA is defined as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (w_1 \times FL_i + w_2 \times NF_i + w_3 \times NN_i - T_i)^2$$

This function shows MSE of the CMIS prediction where w_1 , w_2 and w_3 are the weight factors corresponding to the FL, NF and NN

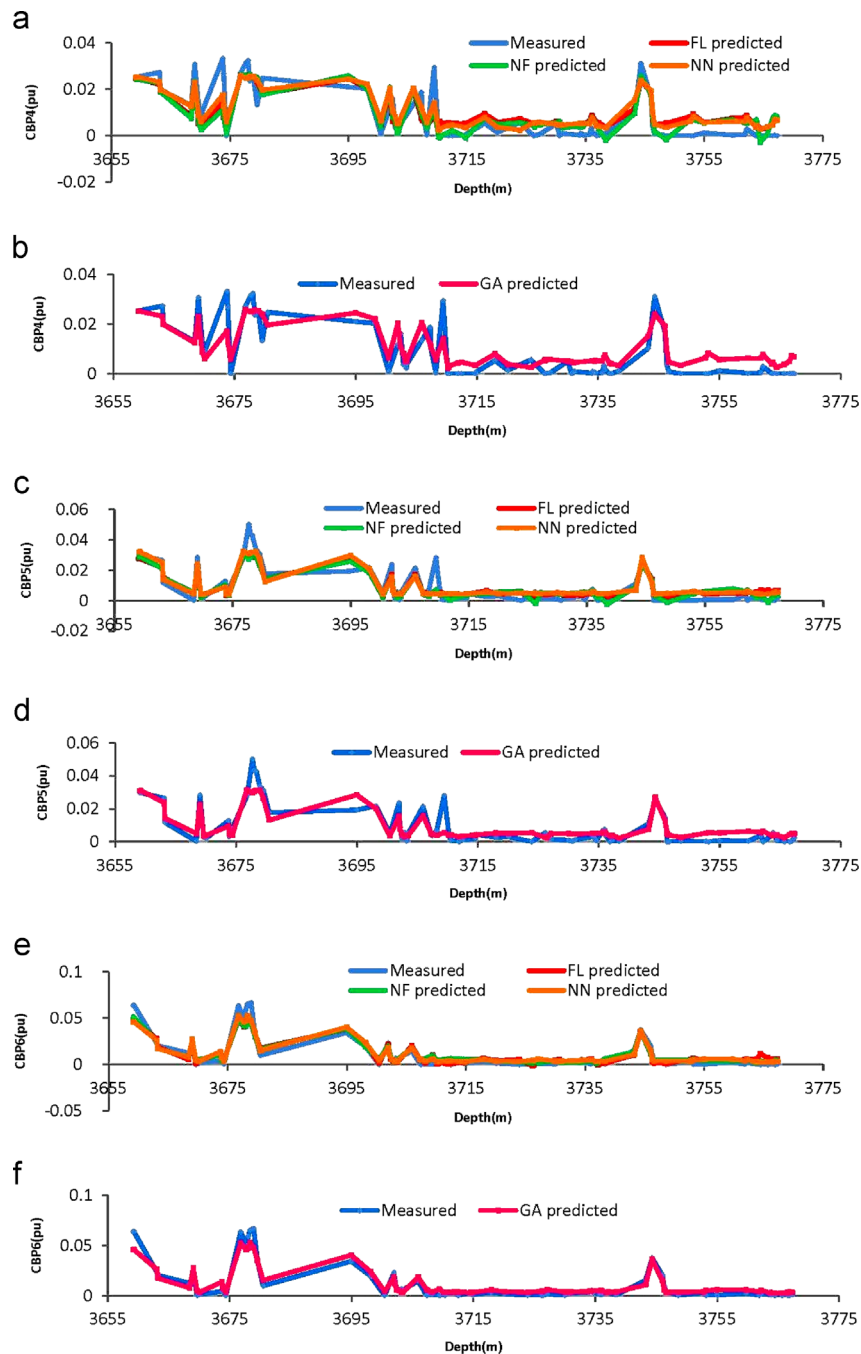


Fig. 5. Comparison between measured and predicted (a, b) CBP4, (c, d) CBP5 and (e, f) CBP6 values in test data versus depth.

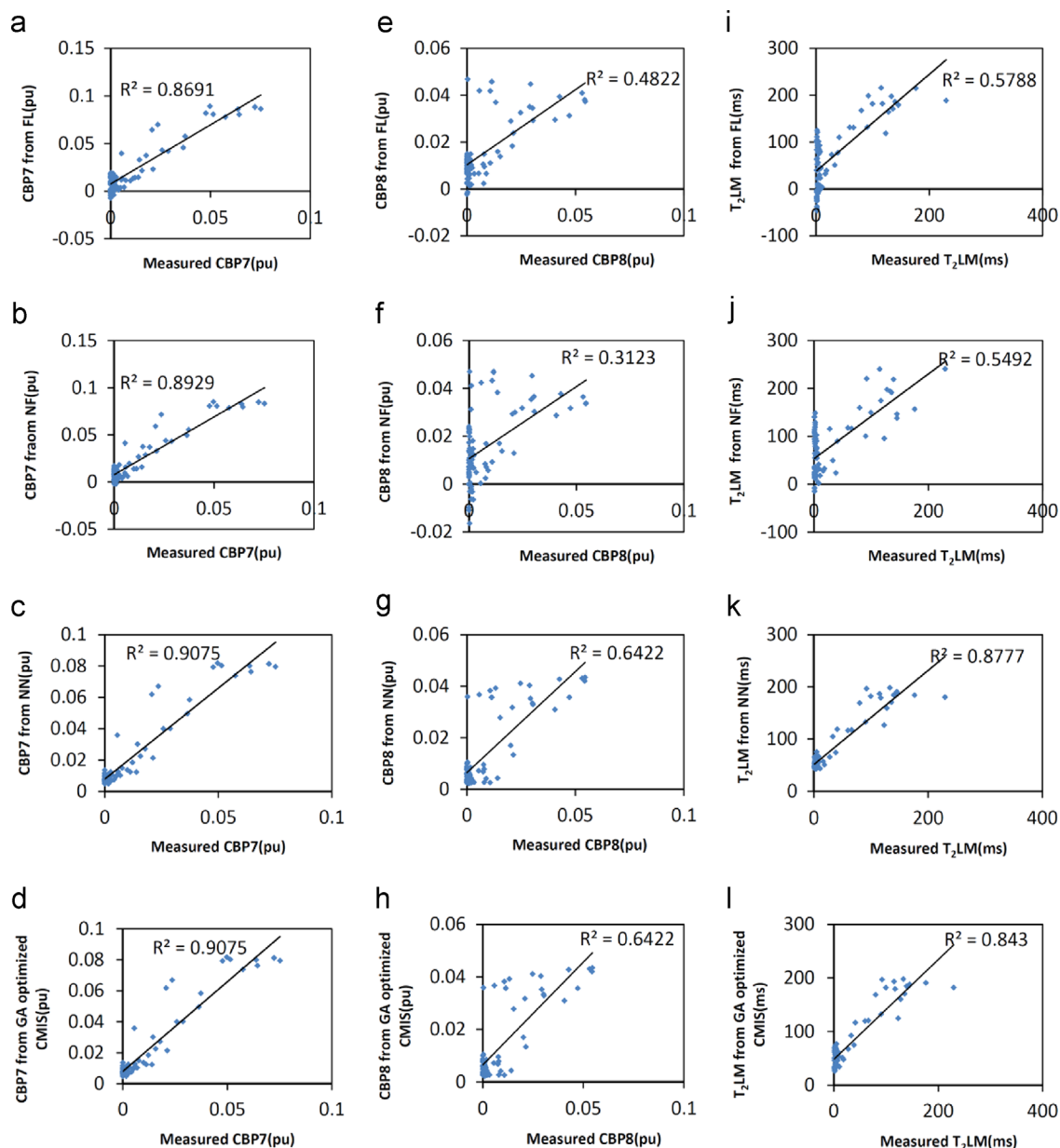


Fig. 6. Cross plots showing correlation coefficients between measured and predicted (a–d) CBP7, (e–h) CBP8 and (i–l) T_2LM values in test data.

predictions, respectively. T_i is the target value (measured CBP or T_2LM using NMR log) and n is the number of training data. A comparison of the performance of the different intelligent models for the CBPs and T_2LM data is shown in Tables 9–11. The correlation coefficient between the NMR measurements and predictions from the intelligent models are displayed in Figs. 2–7. Finally, in the light of the satisfactory results of the CMIS model described in this study the virtual NMR logs are constructed. Graphical comparisons between the measured and estimated T_2 distribution for some T_2 distribution test samples are illustrated in Fig. 8(a–p).

5. Discussion

In this study, the high performance CMIS models were developed for synthesizing the CBP2, CBP1, CBP5, CBP6, CBP4, CBP3, CBP8, CBP7 and T_2LM data. The correlation coefficient between

the measured and estimated NMR curves was more than 0.8 for all models except CBP3 and CBP8. However, in spite of the low correlation values, the average difference between the measured and predicted values or mean squared error is in the same range for all models. For the field studied here, the NN models had superior performance compared with the FL and NF models in all cases. The later statement does not include all NN models constructed from the four training algorithms. However, in some cases the individual NN training algorithms even performed weaker than the FL and NF models. That is, why a higher weight was attributed to the NN models in the weighted averaging optimization. As a result, the optimized models were much closer to the NN models in comparison to the other FL and NF ones. The simple averaging CM was also applied to combine the individual intelligent models. It showed a better performance than either the FL or NF models in all cases. The neuro-fuzzy models performed better than the fuzzy inference system for synthesizing CBP3 through CBP7, while the FL models were more

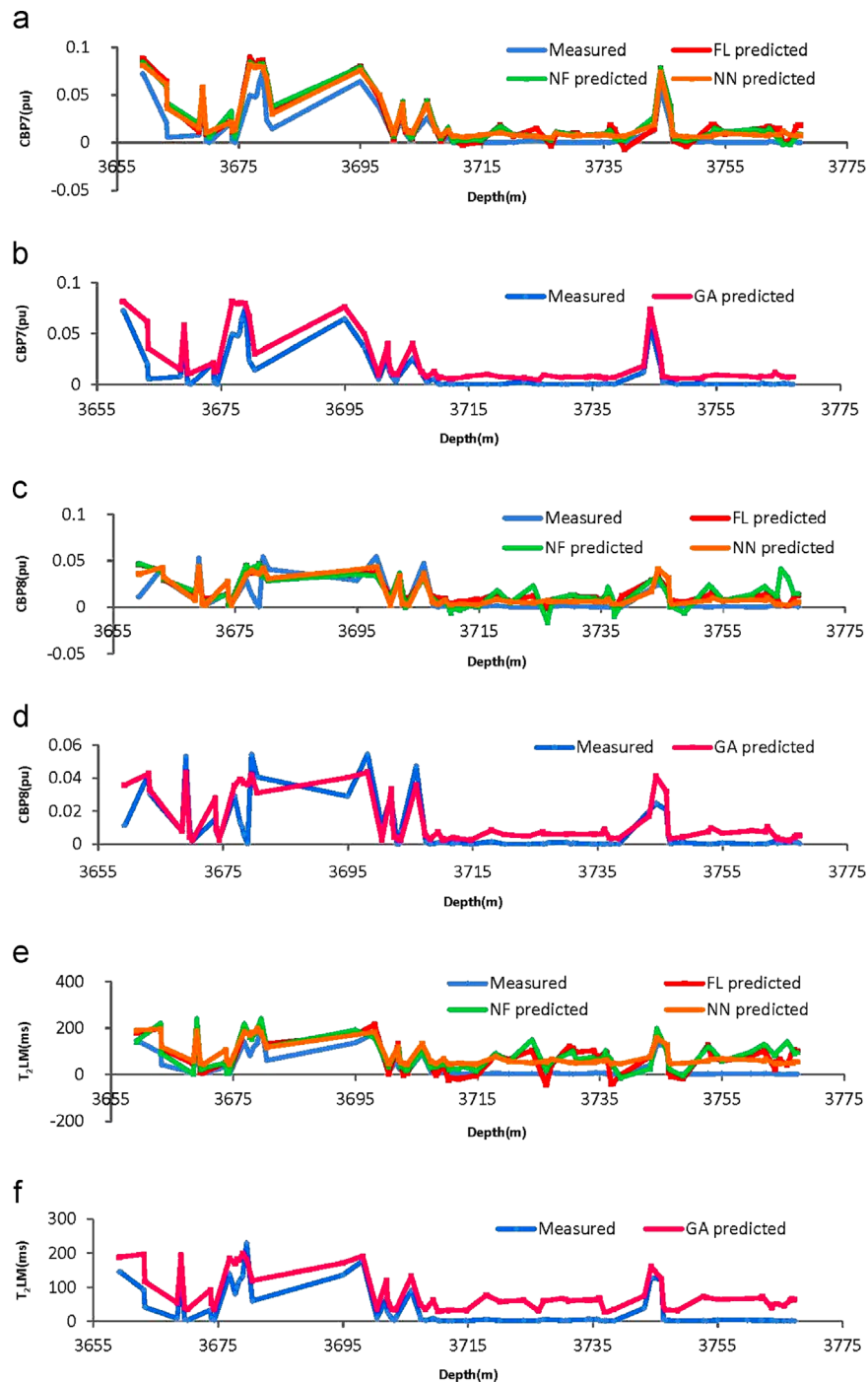


Fig. 7. Comparison between measured and predicted (a, b) CBP7, (c, d) CBP8 and (e, f) T₂LM values in test data versus depth.

successful for estimation of the other CMR bin porosities. By investigating the effects of neural network training parameters on the developed models, it was observed that the number of neurons in the hidden layer did not affect the models' performances considerably. That is why all NN models were developed with 4 neurons in their hidden layer. Even though the number of NF training epochs was greater than 100 in all cases, the optimal NF models were achieved with epochs lower than 50. The CMIS is a cost-effective, time saving and reasonably accurate method that performs better than the individual experts acting alone to estimate the NMR T₂ distribution and T₂ logarithmic mean data from one specific set of conventional well logs.

6. Conclusion

The aim of the present paper was to examine the possibility of synthesizing the T₂ distribution curves from a set of predefined available well log data. The results of this study showed that a committee machine with intelligent systems is capable of predicting the T₂ distribution data and T₂LM values from one specific set of conventional well log data with a reasonable degree of accuracy. Considering the required time, effort and costs of NMR logging in a field-wide extent, this methodology is far superior to practically logging all wells of the field. This work demonstrates that using neutron, density, sonic, lithology, gamma ray and effective

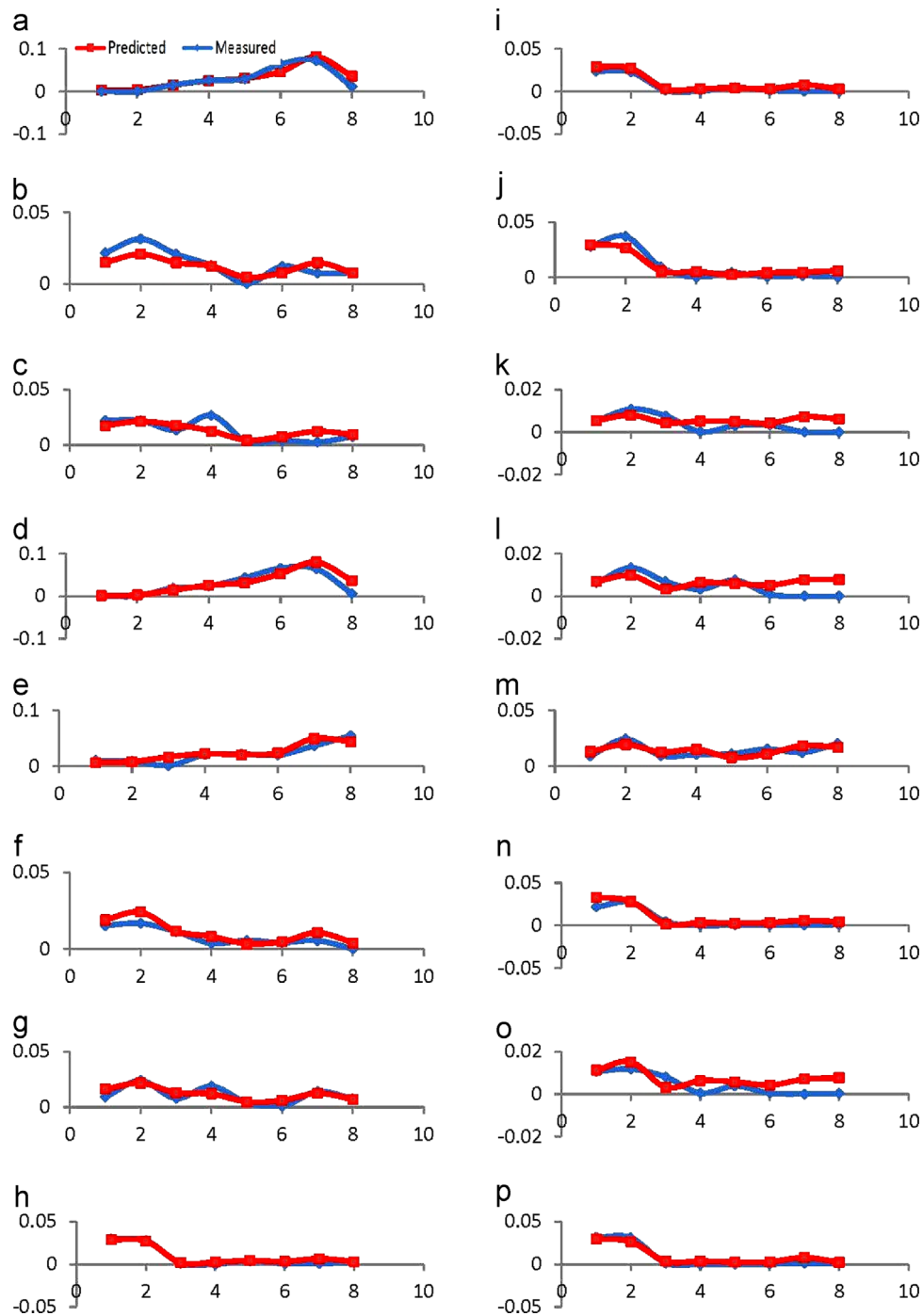


Fig. 8. Comparison between measured and predicted T_2 curves for some test samples (a–p).

porosity logs can offer prediction of T_2 data in a convenient way. A specific set of input well log data with both adequacy and accuracy possesses the NMR related features of the Earth; hence the intelligent models are able to find logical patterns between well logs and NMR T_2 distribution data.

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